

16 REACTOR KINETICS

These cards are required if a space-independent (point) reactor kinetics or a nodal reactor kinetics calculation is desired. These cards may be entered in a new problem or on a restart. If no reactor kinetics data are present in a restart problem, the data will be added; if reactor kinetics data are already present, the data are deleted and replaced by the new data. A complete set of reactor kinetics data must always be entered. Initial conditions are computed the same for new or restart problems; the initial conditions can be obtained from assuming infinite operating time at the input power or from an input power history.

16.1 Reactor Kinetics Type, 30000000

This card is required. Read in by rrkin.

- W1(A) Kinetics type. Enter POINT or DELETE. Enter POINT for the point reactor kinetics option. Enter DELETE in a restart problem if reactor kinetics is to be deleted. No other data are needed if reactor kinetics is being deleted.
- W2(A) Feedback type. Enter SEPARABL, TABLE3, TABLE4, TABLE3A, or TABLE4A. If Word 2 is not entered, a default value is assumed. If the kinetics type is POINT, the default is SEPARABL. If SEPARABL is entered, reactor kinetics feedback due to moderator density, void fraction weighted moderator temperature, and fuel temperature is assumed to be separable, and feedback data are entered on cards 30000501-0899. If SEPARA3D is entered, reactor kinetics feedback due to 3D volume moderator and fuel temperature is assumed. If SEPARA3M is entered, reactor kinetics feedback due to 3D volume moderator and 1D fuel is assumed. If TABLE3, TABLE4, TABLE3A, or TABLE4A is entered, reactivity is obtained from a table defining reactivity as a function of three or four variables using cards 30001001-2999. If TABLE3 or TABLE4 are entered, the variables are moderator density, void fraction weighted moderator temperature, fuel temperature, and boron density. If TABLE3A or TABLE4A is entered, the variables are void fraction, liquid moderator temperature, volume average fuel temperature, and boron concentration. If TABLE3 or TABLE3A is entered, the first three of the variables in one of the sets defined above are used, and if TABLE4 or TABLE4A is entered, all four variables are used.

16.2 Reactor Kinetics Information, 30000001

Read in by rrkin.

- W1(A) Fission product decay type. Enter NO-GAMMA for no fission product decay calculations, GAMMA for standard fission product decay calculations, or GAMMA-AC for fission product decay plus actinide decay calculations.

- W2(R) Total reactor power (W). This is the sum of fission power, fission product decay power, and actinide decay power. Watts are used for both SI and British units. This must be > 0.0 . If SEPARA3D or SEPARA3M is entered, this value is meaningless. Then the total power is given by the 3D input convention.
- W3(R) Initial reactivity (dollars). This quantity must be less than or equal to 0.0. It is recommended that this quantity be less than or equal to $-1.0\text{E-}60$; otherwise, the CPU time may increase dramatically.
- W4(R) Delayed neutron fraction over prompt neutron generation time (s^{-1}).
- W5(R) Fission product yield factor. This is usually 1.0 for best-estimate problems, and 1.2 has been used with ANS73 data for conservative mode problems. The factor 1.0 is assumed if this word is not entered.
- W6(R) ^{239}U yield factor. This is the number of ^{239}U atoms produced per fission times any conservative factor desired. The factor 1.0 is assumed if this word is not entered.
- W7(R) Fissions per initial fissile atom, ψ . Used in factor $G(t) [= 1.0 + (3.24 \cdot 10^{-6} + 5.23 \cdot 10^{-10}t) T^{0.4}\psi]$ to account for neutron capture in fission products when using ANS79-1 or ANS79-3 option. Entering this quantity as a nonzero includes the G factor as part of the decay heat. The factor is not included if this quantity is not entered or is entered as 0. Entering this word as a positive quantity indicates that the equation is to be used for shutdown time up to 10^4 seconds, and the table is to be used from that time on. Entering this word as a negative number indicates that the table is to be used for all shutdown times. Note that there is a discontinuity in $G(t)$ when switching between and equation and the table. The standard indicates that the table can be used for all shutdown times and that would result in a higher neutron absorption capture effect. The magnitude of this quantity if nonzero must be greater than or equal to 1.0 and less than or equal to 3.0.
- W8(R) Reactor operating time T. This quantity is the T in the expression given in W7 above. The unit for this quantity is given in the next word. If not entered or entered as 0, this quantity defaults to 52 wk. This quantity is used only if the power history data in Section 16.7 are not entered. When the power history data are entered, the reactor operating time is obtained from that data. When the power history data are not entered, an infinite operating time is assumed in initializing the decay heat variables, and if the equation form of $G(T)$ is being used, the quantity in this word is used with the shutdown period t set to 0 to determine the G factor at the start of the simulation. The limit for this quantity is $1.2614 \cdot 10^8$ seconds.
- W9(A) Units for W8 above. Must be sec, min, hr, day, wk.

W10(R) Differential boron worth in \$/ppm. This value is normally given in core design reports in units of pcm/ppm. The abbreviation pcm stands for per cent mille or 10^{-5} of $(\Delta k)/k$, while dollars are measured in units of the delayed neutron fraction, $\$ = (\Delta k)/(k\beta)$.

16.3 Fission Product Decay Information, 30000002

This card is optionally entered for POINT problems if W1 of card 30000001 contains GAMMA or GAMMA-AC. If this card is not entered, the Proposed 1973 ANS Standard fission product data are used if default data are used. Read in by rkin.

W1(A) Fission product type. Enter ANS73, ANS79-1, or ANS79-3. If default fission product data are used, ANS73 specifies the Proposed 1973 ANS Standard data, ANS79-1 specifies the 1979 Standard data for ^{235}U , and ANS79-3 specifies the 1979 ANS Standard data for the three isotopes, ^{235}U , ^{238}U , and ^{239}Pu . ANS79-3 also requires that power fractions for each isotope must be entered. ANS94-4 specifies the 1994 ANS Standard data for the four isotopes, ^{235}U , ^{238}U , ^{239}Pu , and ^{241}Pu . ANS79-3 and ANS94-4 also require that power fractions for each isotope must be entered. If fission product data are entered, ANS73 and ANS79-1 specify only one isotope. ANS79-3 specifies three isotopes and ANS94-4 specifies four isotopes. Both ANS79-3 and ANS94-4 require the number of decay heat groups for each isotope be entered.

W2(R) Energy release per fission (MeV/fission). If not entered or 0, the default value of 200 MeV/fission is used.

If using ANS79-3 Data:

W3-W5(R) If ANS79-3 is specified in W1, the fraction of power generated in ^{235}U , ^{238}U , and ^{239}Pu must be entered in these three words. The sum of the fractions must add to 1.0.

W6-W8(I) Number of groups per isotope. If ANS79-3 is entered in W1 and default data are not being used, the number of decay groups for ^{235}U , ^{238}U , and ^{239}Pu must be entered in these words. The number of groups for each isotope must be less than or equal to 50.

If using ANS94-4 Data:

W3-W6(R) If ANS94-4 is specified in W1, the fraction of power generated in ^{235}U , ^{238}U , ^{239}Pu , and ^{241}Pu must be entered in these four words. The sum of the fractions must add to one.

W7-W10(I) Number of groups per isotope. If ANS94-4 is entered in W1 and default data are not being used, the number of decay groups for ^{235}U , ^{238}U , ^{239}Pu , and ^{241}Pu must be entered in these words. The number of groups for each isotope must be less than or equal to 50.

16.4 Delayed Neutron Constants, 30000101-0199

If these cards are missing, constants for the six generally accepted delayed neutron groups are supplied. Otherwise, two numbers for each delay group are entered, one or more pairs per card. Card numbers need not be consecutive. The number of pairs on these cards defines the number of delay groups. Up to 50 delay groups may be entered. Read in by rkin.

W1(R) Delayed neutron precursor yield ratio.

W2(R) Delayed neutron decay constant (s^{-1}).

16.5 Fission Product Decay Constants, 30000201-0299

These cards are not needed if W1 of card 30000001 is NO-GAMMA. If this word is GAMMA or GAMMA-AC, data from these cards or default data are used to define fission product decay. If the cards are missing, data as defined in W1 of card 30000002 are supplied. Up to 50 fission product groups may be entered. Data are entered on cards similarly to cards 30000101-0199. The factor in W5 of card 30000001 is applied to the yield fractions. Read in by rkin.

W1(R) Fission product yield fraction.

W2(R) Fission product decay constant (s^{-1}).

16.6 Actinide Decay Constants, 30000301-0399

These cards are not needed unless W1 of card 30000001 is GAMMA-AC. If GAMMA-AC is entered, data from these cards or default data are used to define actinide decay. If the cards are missing, default data are supplied. Read in by rkin.

W1(R) Energy yield from ^{239}U decay (Mev).

W2(R) Decay constant of ^{239}U (s^{-1}).

W3(R) Energy yield from ^{239}Np (Mev).

W4(R) Decay constant of ^{239}Np (s^{-1}).

16.7 Power History Data, 30000401-0499

If these cards are not present, initial conditions for fission product and actinide groups are for steady-state operation at the power given in W2 of card 30000001. This is equivalent to operation at that power for an infinite time. If these cards are present, the power history consisting of power and time duration is used to determine the fission product and actinide initial conditions. The power from gamma and actinide

decay is assumed to be 0 at the beginning of the first time duration. Data are entered in three- or six-word sets, one or more sets per card. Card numbers need not be consecutive. Read in by rkin.

- W1(R) Reactor power (W). This quantity is the total reactor power, that is, the sum of fission power and decay power, and must be ≥ 0 . If a decay power obtained from the power history exceeds this quantity, the fission power is assumed to be 0.
- W2(R) Time duration. Units are as given in next word. This quantity must be ≥ 0 .
- W3(A) Time duration units. Must be sec, min, hr, day, or wk.
- W4-W7(R) Power fractions. If ANS79-3 or ANS94-4 is entered in W1 of Card 30000002, enter the power fractions for ^{235}U , ^{238}U , and ^{239}Pu in Words 4, 5, and 6. W7 is not entered if using ANS79-3
- If ANS94-4 is entered in W1 of Card 30000002, enter the power fractions for ^{235}U , ^{238}U , ^{239}Pu , and ^{241}Pu in Words 4, 5, 6, and 7.

16.8 Feedback Input

Feedback information for point kinetics information are entered on the following cards. For steady-state computations in which constant power is desired, these cards can be omitted and the feedback reactivity will be 0.

16.8.1 Reactivity Curve or Control Variable Numbers, 30000011-0020

Reactivity (or scram) curves from the general tables (cards 202TTTNN) or control variables that contribute to reactivity feedback are specified on these cards. These cards are not used if there are no references to reactivity contributions from general tables or control variables. Tables and control variables referenced must be defined. No error is indicated if reactivity curves are defined but not referenced on this card, but memory space is wasted. Curve numbers, which are the TTT of the general table card number or control variable number code, are entered one or more per card. Card numbers need not be consecutive. Read in by rkin.

- W1(I) Table or control variable number. Up to 20 numbers may be entered. Numbers from 1-999 indicate general table numbers. Numbers greater than 10000 indicate the control variable whose number is the entered number minus 10000.

16.8.2 Density Reactivity Table, 30000501-0599

This table is required if the SEPARABL option is being used and if cards 30000701-0709 are entered. One or more pairs of numbers are entered to define reactivity as a function of moderator density. Data are entered one or more pairs per card, and card numbers need not be consecutive. Up to 100 pairs

may be entered. The table uses linear interpolation for segments between table search argument values. For search arguments beyond the range of entered data, the end-point values are used. Read in by rrkinp.

W1(R) Moderator density (kg/m^3 , lb/ft^3).

W2(R) Reactivity (dollars).

16.8.3 Doppler Reactivity Table, 30000601-0699

This table is required if the SEPARABL option is being used and if cards 30000801-0899 are entered. One or more pairs of numbers are entered to define Doppler reactivity as a function of volume-averaged fuel temperature. Heat structure composition data cards 1CCCG201-9 need to exclude the gap and the cladding for the volume-average fuel temperatures. Data are entered one or more pairs per card, and card numbers need not be consecutive. Up to 100 pairs may be entered. The table uses linear interpolation for segments between table search argument values. For search arguments beyond the range of entered data, the end-point values are used. Read in by rrkinp.

W1(R) Temperature (K, $^{\circ}\text{F}$).

W2(R) Reactivity (dollars).

16.8.4 Volume Weighting Factors, 30000701-0799

These cards are used only if the SEPARABL option is being used and are omitted if no reactor kinetics feedback from hydrodynamics is present. Each card contains the input for reactivity feedback due to conditions in one or more hydrodynamic volumes. Words 1 and 2 are a volume number and an increment. Words 3 and 4 are the reactivity data for the volume defined by Word 1; Words 5 and 6 are the reactivity data for the volume defined by Word 1 plus Word 2; Words 7 and 8 contain data for the volume defined by Word 1 plus two times Word 2; etc. Each card must contain at least four words. Volumes must be defined by hydrodynamic component data cards, and any volume reactivity data must be defined only once on these cards. Card numbers need not be consecutive.

W1(I) Hydrodynamic volume number.

W2(I) Increment.

W3(R) Weighting factor for density feedback, $W_{\rho i}$. See Volume 1 of this manual for a discussion of the symbols.

W4(R) Water temperature coefficient, a_{W_i} (dollars/K, dollars/ $^{\circ}\text{F}$). As defined in Volume 1, the weighting factor in Word 3 is not applied to this quantity.

16.8.5 Heat Structure Weighting Factors, 30000801-0899

These cards are used only if the SEPARABL option is being used and are omitted if no reactor kinetics feedback from heat structures are present. Each card contains the input for reactivity feedback due to conditions in one or more heat structures representing fueled portions of the reactor. Data are entered in a manner similar to cards 30000701-0799. Read in by rrkinp.

For each heat structure specified on these cards, input on the heat structure data cards 1CCCG2NN must define the fueled region as the region over which the volume-average temperature is computed.

Usually, either Word 3 or 4 is 0.

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| W1(I) | Heat structure number. |
| W2(I) | Increment. |
| W3(R) | Weighting factor for doppler feedback, W_{Fi} . |
| W4(R) | Fuel temperature coefficient, a_{Fi} (dollars/K, dollars/ $^{\circ}$ F). As defined in Volume 1, the weighting factor in Word 3 is not applied to this quantity. |

16.8.6 Volume-Weighting Factors, 30001701-1799

These cards are used only if the TABLE3, TABLE3A, TABLE4, or TABLE4A option is being used. Each card contains the weighting factor for reactivity feedback due to moderator density (void fraction), void weighted moderator temperature (liquid moderator temperature), and spatial boron density (boron concentration) in one or more hydrodynamic volumes. The quantities preceding the quantities within parentheses are used if TABLE3 or TABLE4 has been entered; the quantities within parentheses are used if TABLE3A or TABLE4A has been entered. The same factor is assumed to apply to all three effects, so only one factor is entered for each value. At least three quantities must be entered on each card. The use of the increment field is similar to that in Section 16.8.4. Read in by rrkinp.

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| W1(I) | Hydrodynamic volume number. |
| W2(I) | Increment. |
| W3(R) | Weight factor, W_{pi} . |

16.8.7 Heat Structure Weighting Factors, 30001801-1899

These cards are used only if the TABLE3, TABLE 3A, TABLE4, or TABLE4A option is being used. Each card contains the weighting factor for reactivity feedback due to temperature in one or more heat

structures. At least three quantities must be entered on each card. The use of the increment field is similar to that in Section 16.8.4. Read in by rrkinp.

W1(I) Heat structure number.

W2(I) Increment.

W3(R) Weight factor, W_{Fi} .

16.8.8 Feedback Table Coordinate Data, 300019C1-19C9

If the TABLE3 or TABLE3A option is being used, the feedback table is a function of three variables: moderator density or void fraction ($C = 1$), void weighted moderator temperature or liquid moderator temperature ($C = 2$), and fuel temperature ($C = 3$). If the TABLE4 or TABLE4A option is being used, the feedback table is a function of four variables: the three above and spatial boron density or boron concentration ($C = 4$). Which variables are used depend on the feedback option used (see W2, card 30000000 in Section 16.1). These cards define the coordinates of the table, and table values are entered (on another card set) for each point defined by all combinations of the coordinate values. The table size is the product of the number of coordinate values entered for each variable. At least two coordinate points must be entered, and up to twenty points may be entered for each variable. The table uses multi-dimensional interpolation for values between table coordinate values. For values beyond the range of the entered coordinate values, the end-point coordinate values are used. Coordinate values are entered in increasing magnitude, one or more per card on one or more cards as desired. Card numbers need not be consecutive. The C in the parentheses above defines the C to be used in the card number. Read in by rrkinp.

W1(R) Coordinate value (kg/m^3 , lb/ft^3 for moderator and spatial boron densities; K, $^{\circ}\text{F}$ for moderator and heat structure temperatures; void fractions and boron concentrations are dimensionless).

16.8.9 Feedback Table Data, 30002001-2999

Values defining the table are entered in pairs. The first is a coded number defining the position of the table entry. The second number is the table entry. One or more pairs may be entered on one or more cards as needed. Card numbers need not be consecutive. There is no required ordering for the coded number, but a coded number may be entered only once. Read in by rrkinp.

W1(I) Coded number. The coded number has the form ddmmffbb, where the letter pairs represent coordinate numbers of the independent variables of the table. The dd pair refers to moderator density or void fraction, mm refers to void weighted moderator temperature or liquid moderator temperature, ff refers to heat structure temperature, and bb refers to spatial boron density or boron concentration. The paired numbers range from 00 to one less than the number of coordinate values for that variable. The 00 pair refers to the first

coordinate value. If boron dependence is not included, bb is always 00. All table values must be entered. (A future version may allow gaps that are filled in by interpolation.)

W2(R) Table value.

